Exercise 3

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Introduction

In this analysis, we investigate drug combination responses and gene expression profiles across the NCI-60 panel of human cancer cell lines using statistical and computational approaches introduced during the course. Specifically, we apply machine learning techniques to predict drug combination responses, explore gene expression patterns through dimensionality reduction, perform differential gene expression analysis with respect to TP53 mutation status, and conduct gene set enrichment analysis to identify biological processes associated with these molecular differences. Together, these methods provide a comprehensive framework for understanding drug sensitivity and the transcriptional consequences of key cancer-related mutations.

1. Drug combination response prediction

In this first part of the analysis, we aimed to predict drug combination responses across the NCI-60 cell lines using a Random Forest model. We started by loading the dataset containing information on drug pairs, their concentrations, the corresponding cell line, and the measured response. Then we perform one-hot encoding, which we implemented using the `model.matrix()` function.

To ensure reproducibility of the results, we set a random seed (`set.seed(123)`). This guarantees that the random sampling procedure (used to split the data) produces the same results if the code is run again.

We then created two different training–test split strategies:

1. **Random 80/20 split across all observations**:
   We randomly selected 80% of the rows for training and used the remaining 20% for testing. We trained a Random Forest model with 500 trees on the training set. Then we make predictions using the test set. The final model performance was evaluated using the Pearson correlation between predicted and observed responses.

2. **Cell line–based split (non-overlapping cell lines)**:
   Instead of splitting individual observations, we randomly selected 80% of the unique cell lines for training and used the remaining 20% of entirely unseen cell lines for testing.The same Random Forest settings were used, and performance was again

In the preparation of this report, we used AI tools to support the analysis and presentation of results. Specifically, ChatGPT was used to review R code, clarify the function of commands, understand parameter settings, and assist in debugging errors. Gemini was used to improve the styling and readability of plots and figure legends. All interpretations, conclusions, and final decisions remain the authors' own.

assessed using the Pearson correlation.

Finally, these are the results of the Pearson Correlation for both models:

Pearson correlation (All data): 0.9444678
Pearson correlation (Cell-line split): 0.7577354

What we observe is consistent with theoretical expectations: the first scenario yields higher performance because the model may encounter similar cell line profiles in both training and test sets. In contrast, the second scenario is more challenging, as the model must predict responses for completely unseen cell lines, requiring stronger generalization ability. Performance in this setting could potentially be improved by incorporating additional biological features (e.g., gene expression profiles), applying feature selection in the random forest model, tuning hyperparameters using a validation set, or increasing model complexity.

2. Gene expression data exploration

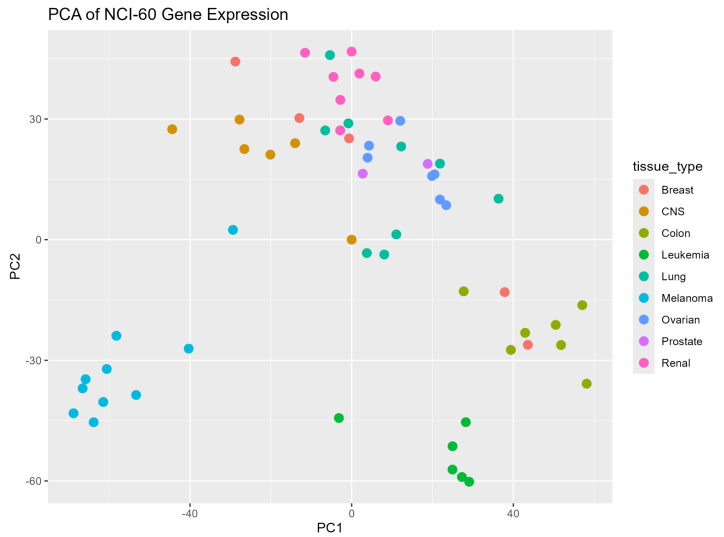
In this part of the analysis, we explored the gene expression profiles of the NCI-60 cell lines to understand their global transcriptional variability. We began by loading the normalized gene expression dataset together with the corresponding metadata, which includes information such as the tissue of origin for each cell line.

First of all we checked for missing values in the expression matrix. And we imputed all the missing values with the mean of each variable.

We then performed Principal Component Analysis (PCA) using the `prcomp()` function on the expression matrix. We calculated the proportion of variance explained by each principal component and extracted the variance explained by the first two components (PC1 and PC2). These are the results:

Variance explained by PC1: 11.29618 %
Variance explained by PC2: 10.28127 %

To visualize the results, we created a scatter plot of the samples along PC1 and PC2.



By examining the PCA plot, we can observe well-defined clusters corresponding to Melanoma, Leukemia, and Colon cell lines.

3. Differential gene expression analysis

We investigated whether TP53 mutation status is associated with systematic differences in gene expression across the NCI-60 cell lines.

We first extracted the TP53 mutation status from the metadata. After we constructed a design matrix using the `model.matrix()` function. We renamed the columns of the design matrix to make the comparison clear and interpretable.

Next, we defined a contrast matrix using `makeContrasts()` to specify the exact comparison of interest: TP53-mutated versus TP53-wildtype cell lines.

Finally we fitted a linear model to each gene using `lmFit()` applying the empirical Bayes moderation.

To account for multiple hypothesis testing across thousands of genes, we adjusted p-values using the Benjamini–Hochberg (BH) procedure. Genes with adjusted p-values below 0.05 were considered statistically significant. Here the list of significantly differentially expressed genes:

Gene	logFC	adj.P.Val
MDM2	1.174026	1.52E-05
CDKN1A	1.780371	1.52E-05

EDA2R	1.172437	1.73E-05
FDXR	0.898786	1.14E-04
PTCHD4	1.040817	3.62E-04
ANKRD20A12P	0.930168	0.001376
RPS27L	1.037150	0.001554
BAX	0.888336	0.001986
RRM2B	0.790896	0.003220
TMEM5	0.783316	0.003582
ZNF878	-0.616087	0.006414
DDB2	0.809526	0.006470
XPC	0.609035	0.007733
ZNF266	-0.800829	0.012894
GDF15	2.293752	0.015741
SPATA18	0.819284	0.019143
FTCDNL1	0.821380	0.019143
SESN1	0.880444	0.019143
SNX18P2	0.498621	0.040105
DCP1B	0.736452	0.040105
MIR3189	0.785567	0.040105

Then we created a volcano plot to visualize the results.

From the volcano plot we can assess that the genes represented by points at the top of the plot exhibit very low p-values, indicating high statistical significance. These genes represent prime candidates for further investigation, as they display consistent and reliable changes in expression between TP53-mutated and wild-type cells.

3. Gene set enrichment analysis (GSEA)

To identify biological processes that are significantly associated with differentially expressed genes, we performed a gene set enrichment analysis using a hypergeometric test. For each GO term, we tested the null hypothesis H0: “Differentially expressed genes are independent of the biological process”.

First, we loaded a collection of gene sets corresponding to GO terms. For each term, we tested whether the number of differentially expressed genes observed in the set is higher than expected by chance. Then the p-values obtained were then adjusted for multiple testing using the Benjamini-Hochberg method. We listed the GO term with an adjusted p-value below 0.05:

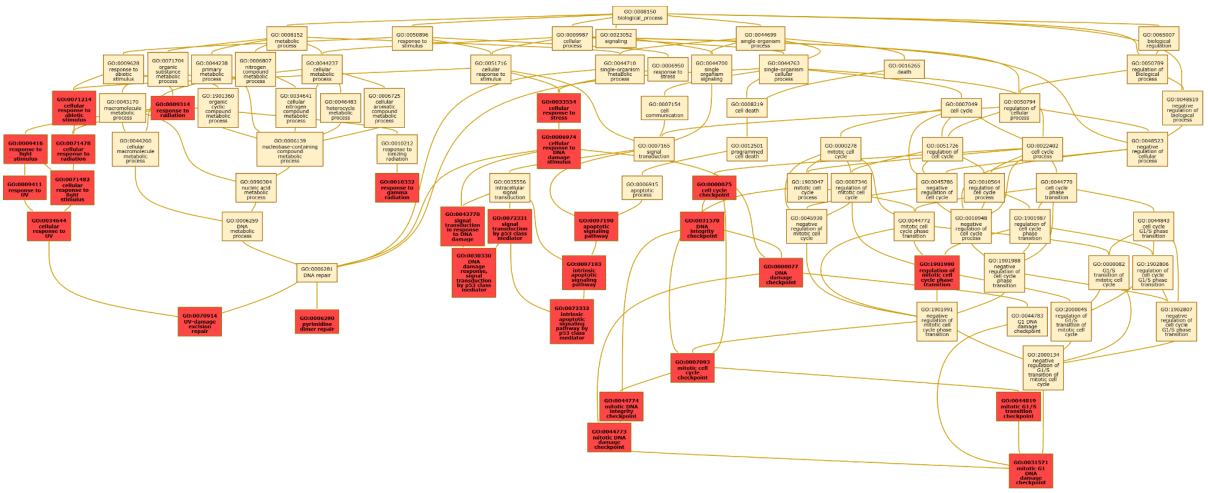
GO Term	Adjusted P-value
GO:0072332	8.14 x 10^-7
GO:0072331	3.21 x 10^-5

GO:0034644	4.96 x 10^-5
GO:0071482	1.37 x 10^-4
GO:0009411	3.61 x 10^-4
GO:0097193	3.61 x 10^-4
GO:0006974	5.04 x 10^-4
GO:0071478	5.04 x 10^-4
GO:0044773	7.91 x 10^-4
GO:0044774	8.81 x 10^-4
GO:0033554	1.45 x 10^-3
GO:0031571	1.83 x 10^-3
GO:0044819	1.83 x 10^-3
GO:0000077	2.98 x 10^-3
GO:0031570	3.76 x 10^-3

GO:0009416	3.95 x 10^-3
GO:0007093	4.00 x 10^-3
GO:0071214	4.62 x 10^-3
GO:0104004	4.62 x 10^-3
GO:0097190	4.96 x 10^-3
GO:1901990	4.96 x 10^-3
GO:0010332	5.56 x 10^-3
GO:0042770	9.57 x 10^-3
GO:0000075	9.80 x 10^-3

This approach allows us to identify biological processes that are disproportionately affected by TP53 mutations in a clear and systematic way, providing insights into pathways potentially involved in cancer-related mechanisms such as apoptosis, DNA repair, or cell cycle regulation.

Visualization with the GOView tool:



Here the 3 GO terms that are related to established cancer biology:

GO:0006281
GO:0006915
GO:0051726

We performed the GSEA using the runGSA from the piano package. We used the gene set collection provided in "*data/gene set collection.RData*". Additionally we prepared the log fold change (logFC) results from the differential gene expression analysis.
We registered 22 GO terms enriched with up-regulated genes and 24 GO terms enriched with down-regulated genes.
Here below the list:

Up Terms:

GO Term	Genes (tot)	Stat (dist.dir)	Genes (up)	Genes (down)
GO:0006582	25	0.84	19	6
GO:0010623	22	0.74	13	9
GO:0019748	50	0.75	31	19

GO:0030318	24	0.81	20	4
GO:0032438	38	0.74	31	7
GO:0033059	61	0.67	46	15
GO:0042438	23	0.84	18	5
GO:0042440	73	0.63	52	21
GO:0042771	41	0.63	24	17
GO:0043455	16	0.82	11	5
GO:0043473	109	0.67	72	37
GO:0044550	25	0.82	19	6
GO:0046148	53	0.66	39	14
GO:0046189	46	0.73	27	19
GO:0048021	16	0.82	11	5
GO:0048023	10	0.87	7	3

GO:0048066	48	0.75	35	13
GO:0048753	39	0.75	32	7
GO:0050931	34	0.78	27	7
GO:0072332	78	0.57	48	30
GO:1900376	16	0.82	11	5
GO:1900378	10	0.87	7	3

Down Terms:

GO Term	Genes (tot)	Stat (dist.dir)	Genes (up)	Genes (down)
GO:0006139	5069	-0.21	2361	2708
GO:0006351	3201	-0.25	1441	1760
GO:0006357	2350	-0.29	1011	1339
GO:0006366	2439	-0.28	1061	1378
GO:0006725	5272	-0.20	2464	2808

GO:0009889	4712	-0.22	2162	2550
GO:0010467	5657	-0.20	2664	2993
GO:0010468	4457	-0.23	2016	2441
GO:0010556	4574	-0.23	2086	2488
GO:0010669	30	-0.64	11	19
GO:0016070	4155	-0.23	1918	2237
GO:0018130	3653	-0.24	1685	1968
GO:0019219	3638	-0.24	1637	2001
GO:0019438	3666	-0.23	1691	1975
GO:0031326	4684	-0.22	2144	2540
GO:0032774	3226	-0.25	1458	1768
GO:0034641	5639	-0.20	2674	2965
GO:0034654	3583	-0.24	1642	1941

GO:0044271	4332	-0.22	2053	2279
GO:0046483	5222	-0.21	2439	2783
GO:0051252	3392	-0.25	1519	1873
GO:0090304	4629	-0.22	2139	2490
GO:1901362	3803	-0.23	1749	2054
GO:2001141	3118	-0.25	1393	1725

Finally we comparde the results from the hypergeometric test and from the gene set enrichment analysis (GSEA). There was only one single term that overlapped: GO: 0072332.